

Alexander Hay

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Education

Northwestern University M.S. Robotics	Evanston, IL 2020
Iowa State University B.S. Mechanical Engineering	Ames, IA 2014

Skillset

Product Development, Modeling and Design, Complex Problem-Solving, Data Science, Management, Leadership

Experience

Translating complex problems and data into actionable solutions

Naxos Lab LLC - Montreal, QC <i>CTO & Lead Engineer</i>	Mar 2022 - Jan 2023
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- Collaborated with US and Canadian universities designing novel laboratory enrichment devices
- Improved quality and reduced production time by optimizing model and machine parameters
- Drafted information deliverables, including charts, models, BoMs, budgets, and reports for stakeholders
- Implemented ISO guided chain of visibility, traceability, and accountability

Universite de Montreal - Montreal, QC <i>Lab Manager - Research Associate</i>	Jan 2021 - Feb 2022
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- Managed an interdisciplinary team spanning neuroscience, data science, mechanical engineering
- Designed, refined, and developed series of novel enrichment devices for NHP
- Rehabilitated NHP for reintroduction into laboratory environment, encouraging freedom of use
- Developed and validated models translating neural and biomechanical signals into real-time system control
- Applied statistical modeling, machine learning, and feature engineering to interpret high-dimensional data
- Coordinated with inter-university departments, including facility, finance, and animalerie
- Implemented lab equipment management program, complying with university guidelines

Northwestern University - Chicago, IL <i>Graduate Researcher - Robotic and Neural Systems</i>	Aug 2019 - Dec 2020
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- Led a cross-functional team through challenges in object detection, motion planning, and robotic manipulation, translating model results into actionable design adjustments
- Applied statistical modeling in Python to build and optimize an AI agent for complex decision-making (cribbage)
- Engineered data-driven motion models for robotic systems, leveraging AI algorithms to predict and refine complex movement dynamics
- Visualized high-dimension, inter-neuron communication using quantitative image analysis

Pearce Services - Des Moines, IA <i>Design Engineer</i>	Apr 2016 - Aug 2019
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- Facilitated telecom infrastructure development using GIS software for major metro ISPs and carriers
- Managed multiple projects accordingly with dynamic priorities and clients
- Forecasted construction, labor, and material budget, dispatched engineers
- Composed BoM, drawings, and layouts for clients in final design report

Full work history and references available upon request